EA-MT Entity-Aware Machine Translation in English to Chinese

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Jou-Yi Lee
jlee1039@ucsc.edu

Abstract

The primary objective of this project is to tackle the challenge of translating a given input sentence from the source language, English, into the target language, Chinese. This task focuses on addressing the accurate translation of named entities, which may be rare, ambiguous, or unknown to traditional machine translation systems.

The goal is to develop a machine translation system capable of effectively handling such named entities within input sentences, ensuring that their translation into the target language is precise.

1 Introduction

Named entity translation presents a unique set of challenges in machine translation systems due to their reliance on contextual, structural, and linguistic nuances. Ambiguity in context is a pervasive issue, as entities often depend on surrounding text for proper interpretation. For instance, the term "*Apple*" may refer to the fruit or the technology company, and distinguishing between these interpretations becomes particularly challenging when the contextual clues are limited or absent.

Another complexity arises from multi-word entities, such as names of organizations, products, or events, which require precise handling to maintain semantic accuracy. Literal translations of such entities often lead to awkward or incorrect results. For example, translating "*Department of Justice*" word-for-word fails to convey the appropriate equivalent term in the target language, thus diminishing the overall quality of the translation.

Cross-lingual consistency further complicates the translation process, as entities may have multiple valid representations in the target language. Maintaining consistency across translations is critical to ensuring that the intended meaning is preserved in various linguistic contexts.

To address these challenges, this study employs the **IWSLT 2017 English-to-Chinese** dataset, focusing on named entity translation. Named entities in the dataset are first identified using **SpaCy**, a robust named entity recognition (NER) tool. These recognized entities are then incorporated into a machine translation pipeline based on the **mBERT** architecture, leveraging pre-trained and fine-tuned weights to enhance the translation accuracy of complex and ambiguous entities. This methodology seeks to establish a robust framework for improving named entity translation, ensuring semantic and contextual fidelity in cross-lingual tasks.

2 IWSLT Dataset

The International Conference on Spoken Language Translation (IWSLT) 2017 dataset involves English-to-Chinese translations and provides a valuable resource for real-world conversational examples. For instance:

> *"Thank you so much, Chris. And it's truly a great honor to have the opportunity to come to this stage twice; I'm extremely grateful."*
>
> "非常谢谢，克里斯。的确非常荣幸能有第二次站在这个台上的机会，我真是非常感激。"

In this dataset, English sentence lengths are more dispersed compared to Chinese sentences. Chinese sentence lengths are concentrated

in shorter intervals because Chinese is a character-based language and typically uses fewer words.

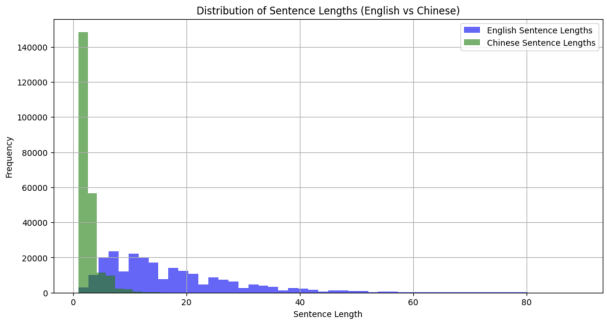


图 1: Distribution of Sentence Lengths (English vs. Chinese)

2.1 Linguistic Structural Differences

- **Character-Based vs. Word-Based Languages:**

  Chinese, as a character-based language, treats each character as the smallest semantic unit. In contrast, English, as a word-based language, requires additional auxiliary words (e.g., articles, prepositions, auxiliary verbs) to form complete sentences. Common examples of auxiliary words in English include articles (*"the"*), prepositions (*"in"*), and auxiliary verbs (*"is"*).

  For instance:

  – English: *"The dog is on the mat."* (6 words)

  – Chinese: "狗在墊子上。" (5 characters)

  This demonstrates that Chinese often uses fewer units to convey the same meaning, while English relies on additional grammatical components to structure sentences.

- **Differences in Word Order and Syntax:**

  English generally follows a fixed Subject-Verb-Object (SVO) structure, which makes sentence construction predictable and uniform. On the other hand, Chinese exhibits more flexibility in word order, allowing sentence structure to vary depending on the intended emphasis. However, this flexibility in Chinese may lead to alignment challenges during translation, especially when attempting to map flexible Chinese structures to the fixed SVO patterns of English.

  For example:

  – English: *"The red book on the table belongs to me."*

  – Chinese: "桌上的紅色書是我的。" (Reordered emphasis on *"the red book on the table"*)

  Furthermore, Chinese allows for reordering of elements to emphasize different parts of a sentence:

  – Standard: "我喜歡這本書。" (Subject-first: *"I like this book."*)

  – Emphasis: "這本書我喜歡。" (Emphasizing *"this book"*)

These structural differences pose challenges for machine translation models. While English requires adherence to strict grammatical structures, Chinese translation must account for its compact nature and flexible word order. Misalignment of sentence structures during translation can result in semantic or syntactic inaccuracies. Models must, therefore, be designed to accommodate the unique linguistic characteristics of both languages to achieve high-quality translations.

IWSLT English Word Cloud of Tokens in the Dataset

The word cloud illustrates the most common tokens within the IWSLT English-to-Chinese translation dataset. As expected, function words such as "the," "and," "to," "of," and "a" are prominently featured due to their frequent usage in the English language. This observation aligns with Zipf's Law, which states that in most natural language texts, function words tend to appear far more frequently than semantic words.

Furthermore, the word cloud highlights tokens such as "can," "they," and "this," which suggest the presence of conversational or descriptive text within the dataset. Although function words dominate the visualization, less frequent tokens like "what," "years," and "think" indicate potential narrative or explanatory content.

图 2: Word Cloud of Tokens in the IWSLT Dataset

IWSLT Chinese Word Cloud of Tokens in the Dataset

As shown in the visualization, high-frequency tokens such as “的,” “是,” “了,” “在,” and “有” dominate the dataset. These words are function words in Chinese and play a key role in sentence structure, similar to their English counterparts like “the,” “and,” or “to.” This observation aligns with Zipf’s Law, which posits that functional words frequently outnumber semantic words in natural language texts.

The visualization also highlights additional words like “我们,” “他们,” and “自己,” which indicate conversational or narrative contexts within the dataset. While function words are predominant, other tokens such as “因为,” “可以,” and “为什么” suggest the presence of explanatory or argumentative content.

图 3: Word Cloud of Tokens in the IWSLT Dataset (Chinese)

The analysis reveals that both character-based and word-based languages follow Zipf’s Law. This law highlights that in natural language, a small number of high-frequency tokens dominate the dataset, while the majority of tokens appear infrequently. The word cloud further reflects the diversity of the IWSLT dataset, suggesting that it likely contains a mix of narrative, conversational, and explanatory text types. The analysis reveals that both character-based and word-based languages follow Zipf’s Law. This law highlights that in natural language, a small number of high-frequency tokens dominate the dataset, while the majority of tokens appear infrequently. This visualization indicates that the dataset includes diverse text genres, such as narrative, conversational, and explanatory elements. It offers a preliminary understanding of the dataset’s composition and variety. It can guide preprocessing strategies, such as removing stopwords (e.g., “The", "a",“的,” “是,” “了,”) to focus on more semantically meaningful tokens for downstream natural language processing tasks.

In the Figure(4) we can see the complex sentence type appears most frequently. Complex sentences are typically sentences with one independent clause and one or more dependent clauses, often involving subordinating conjunctions like "because," "although," or "since.". Compound sentences are the second most frequent type it consist of two or more independent clauses connected by coordinating conjunctions such as "and," "but," or "or."

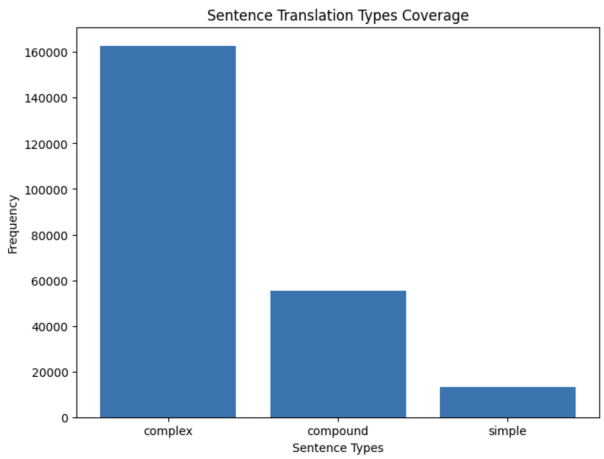
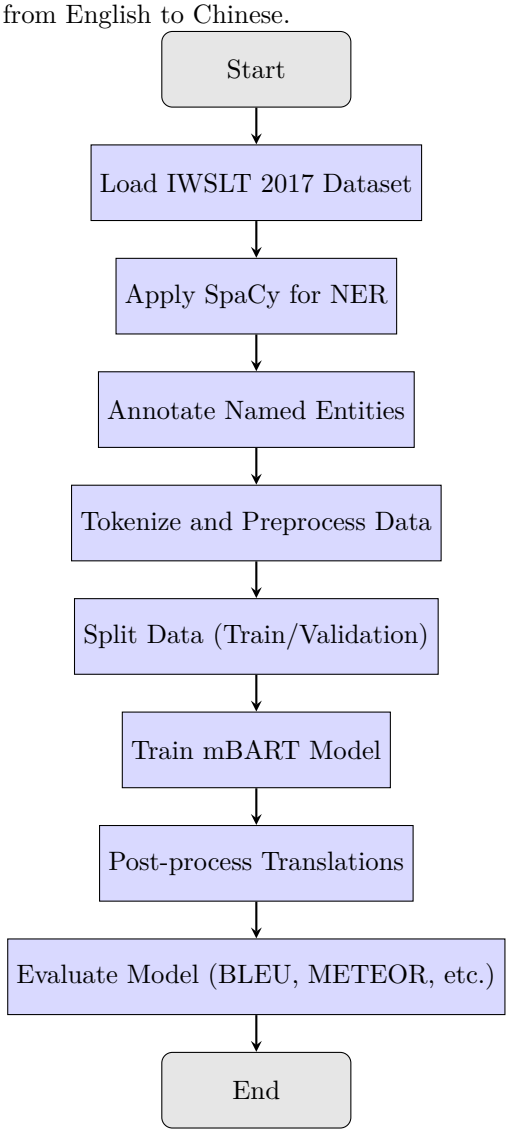


图 4: The Translation Coverage Types in IWSLT Dataset

The high frequency of complex sentences mean the translation model to handle subordinate clauses effectively.

3 Methodology - Flowchart

The process of integrating entity-aware translation is divided into several steps to manage the complexities of accurately translating named entities from English to Chinese.



In this project, I utilized SpaCy to extract named entities from English sentences and customized the entity annotation format (e.g., «`ENTITY:TEXT`»). This approach enhances the model's ability to recognize and handle named entities effectively. Next, the pre-annotated entity data was used as input for the mBART model. During training, the mBART model was fine-tuned with a focus on high-frequency entities such as `CARDINAL`, `DATE`, and `PERSON`. After translation, the entity annotations in the translated results were restored to readable text, ensuring that the output was both fluent and accurate. Multiple evaluation metrics, including METEOR and ROUGE, were employed to assess the model's performance in translating entities and maintaining contextual accuracy.

My strategy successfully combined the advantages of entity-aware annotations and a multilingual translation model, addressing the core challenge of named entity translation in English-to-Chinese tasks. This method significantly improved the model's understanding and translation accuracy of entities, achieving a strong translation score (METEOR 0.6649). In the following sections, I will provide a step-by-step explanation of each stage of the process.

4 Data Preprocessing

In this study, the **IWSLT** 2017 English-to-Chinese translation dataset was utilized, with a focus on addressing the challenge of translating named entities. To enhance the model's ability to process named entities, additional steps were taken to improve entity recognition. Specifically, **SpaCy's** Transformer-based English named entity recognition model (`en_core_web_trf`) was employed to identify entities within the source sentences. The identified entities were annotated in the format «`{ent.label}:{ent.text}>` , for example, «`ORG:Apple`». This annotation allows the model to recognize these key segments during training and inference. This preprocessing step significantly enhances the model's awareness of named entities, thereby optimizing translation quality.

5 Named Entity Recognition

Named entities can be conceptualized as specific instances of broader entity categories (for example, "Apple" is an instance of the category `Organization`). In the domain of information extraction, a named entity refers to a real-world object or concept that can be identified by a proper name, such as a person, location, organization, or product. These entities can exist either in physical form or in abstract form. Recognizing named entities is crucial for tasks such as machine translation, where the accurate identification and handling of named entities can significantly enhance translation precision.

5.1 SpaCy: `en_core_web_trf`

SpaCy provides pre-trained models for multiple languages. For this study, I selected the `en_core_web_trf` model to identify all named entities in the IWSLT dataset. This model leverages the Transformer architecture, which significantly improves the accuracy of named entity recognition, particularly in contexts with complex linguistic structures.

5.2 Named Entity Tags and Descriptions

The `en_core_web_trf` model supports the following entity tags:

Using SpaCy, I employed the `.label_` attribute to extract named entities and implemented a custom Python function, `mark_entities()`, to format the extracted entities into a structured representation of the form «`{ent.label}:{ent.text}`>. This explicit tagging approach enhances the model's ability to recognize and handle named entities during the translation process. By explicitly marking entities, the translation model gains a better understanding of their semantic significance, reducing the likelihood of misinterpretation or loss of meaning during translation.

5.3 Named Entity Tags Distributiion in IWSLT Dataset

In Figure(5), it can be observed that `CARDINAL` (numerical data) and `DATE` (dates) are the most frequently occurring entity types. This highlights the importance of numbers and dates in the dataset, such as references to years, statistics, or expressions of time. `PERSON` (names of people) and `GPE` (geopolitical entities like cities or countries) closely follow, emphasizing their critical role in improving Entity-Aware Machine Translation. These frequent entity types are vital for the model to learn and recognize entities effectively, which significantly improves the translation's accuracy.

Tag	Description	Example
PERSON	A person's name	"Barack Obama"
NORP	Nationalities, religious groups, or political groups	"American"
FAC	Buildings, landmarks, or facilities	"Eiffel Tower"
ORG	Organizations, companies, or institutions	"Google"
GPE	Countries, cities, or geopolitical regions	"France"
LOC	Non-GPE locations, such as mountain ranges or bodies of water	"Mount Everest"
PRODUCT	Objects, vehicles, or products	"iPhone"
EVENT	Named events	"Olympics"
WORK_OF_ART	Titles of books, songs, or artworks	"Mona Lisa"
LAW	Legal documents or clauses	"First Amendment"
LANGUAGE	Languages	"English"
DATE	Dates or time periods	"2024-01-01", "Wednesday"
TIME	Specific times	"2 PM", "noon"
PERCENT	Percentage expressions	"50%"
MONEY	Monetary amounts	"$20"
QUANTITY	Measurements	"10 kg"
ORDINAL	Ordinal numbers	"first", "2nd"
CARDINAL	Cardinal numbers	"one", "100"

表 1: Named entity tags, their descriptions, and examples in `en_core_web_trf`.

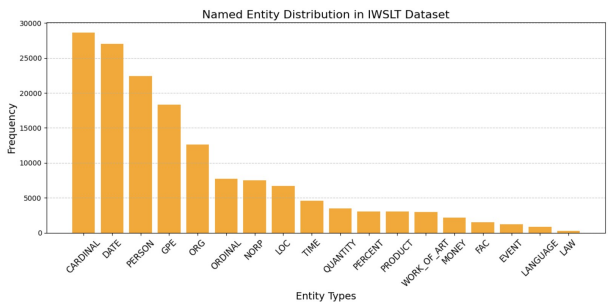


图 5: NER Bar Chart in the IWSLT Dataset

In Figure(6), the distribution of entity tags identified from the IWSLT dataset is shown. Most entities fall into ten primary categories. This further demonstrates that optimizing the model's Entity-Aware translation capabilities requires focused training on high-frequency types, such as `CARDINAL` and `DATE`.

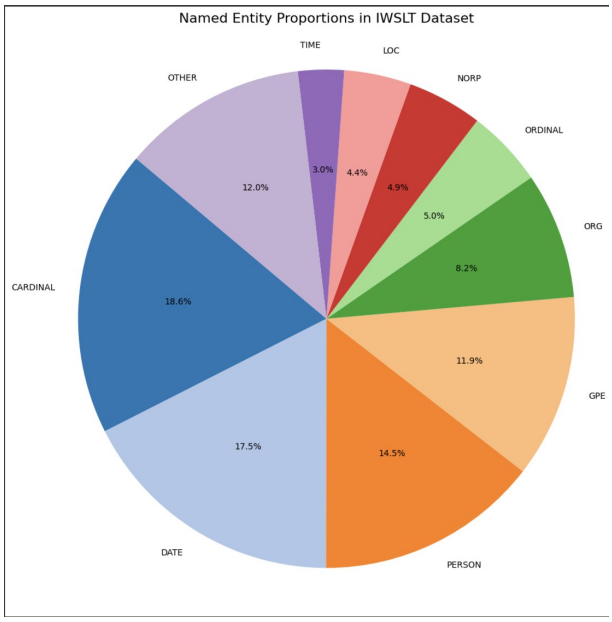


图 6: NER Pie Chart in the IWSLT Dataset

6 Model Design — mBART

Traditional language models, such as ELMo and OpenAI GPT, typically adopt unidirectional or shallow bidirectional architectures, which limit their ability to fully leverage bidirectional context. For example, GPT uses a constrained self-attention mechanism, where each token can only attend to its left-side context.

The mBART model is an encoder-decoder architecture specifically designed for generation tasks. In this study, we utilize a multilingual translation model developed by Facebook AI (now Meta AI), based on mBART (Multilingual BART). The architecture employs a multi-layer Transformer-based encoder-decoder structure, with noise injection (e.g., sentence shuffling or token masking) applied during pretraining.

This translation system is built upon the mBART model (`mbart-large-50-many-to-many-mmt`), a pre-trained Transformer-based sequence-to-sequence model designed for multilingual tasks. The tokenizer of the mBART model is customized to correctly recognize and process inputs with entity markers. By fine-tuning the preprocessed dataset, the model leverages the additional contextual information provided by named entity annotations, thereby improving translation performance.

6.1 mbart-large-50-many-to-many-mmt

Pretrained multilingual models, such as mBERT and mBART, have demonstrated significant potential in handling multilingual tasks. BERT (Bidirectional Encoder Representations from Transformers) is a model developed by Google, and its multilingual version, mBERT (Multilingual BERT), utilizes a Masked Language Model (MLM) approach, primarily for understanding tasks such as text classification and question answering. BERT is trained using a bidirectional self-attention mechanism. In contrast, mBART is an unsupervised translation model designed for multilingual sequence-to-sequence generation tasks, based on denoising autoencoder pretraining.

The **mbart-large-50-many-to-many-mmt** model is fine-tuned for Multilingual Machine Translation (MMT). It can directly translate between 25 languages, including English (En) and Chinese (Zh).

According to the paper *Multilingual Denoising Pre-training for Neural Machine Translation*, **Facebook/mBART** achieved up to a 12 BLEU score improvement for low-resource languages and a 9.5 BLEU improvement for unsupervised translation tasks. This highlights the powerful capabilities of multilingual denoising pre-training in translation tasks. The paper *Multilingual Translation with Extensible Multilingual Pretraining and Fine-tuning* further explains that the mBART model has been extended to support 50 languages (compared to the original mBART, which supported only 25). The **mbart-large-50-many-to-many-mmt** model was pre-trained using the ML50 benchmark dataset, which includes 50 languages ranging from high-resource languages (e.g., German, French) to medium-resource languages (e.g., Romanian, Malayalam) and low-resource languages (e.g.,

Gujarati, Mongolian).

The paper *Multilingual Denoising Pre-training for Neural Machine Translation* also elaborates on the architecture of **mBART**, which uses a Transformer encoder-decoder structure with 12 layers for both the encoder and decoder, a hidden dimension of 1024, and a total of 680 million parameters. The model outperforms randomly initialized models significantly, even when parallel corpora are reduced to 10% of the original data. The paper also tested the model on datasets such as TED Talks (IWSLT) and Europarl, covering both high-resource and low-resource language pairs. **mBART** demonstrated outstanding performance in unsupervised machine translation tasks, especially for low-resource languages.

mBART50 provides three configurations:

- **Many-to-One (N1):** mBART50 outperforms mBART25 with a 2.5 BLEU improvement.
- **One-to-Many (1N):** mBART50 achieves a 5.7 to 7 BLEU improvement over mBART25.
- **Many-to-Many (NN):** mBART50 delivers a 0.8 BLEU improvement compared to mBART25.

7 EA-MT: Entity-Aware Machine Translation from English to Chinese

EA-MT leverages SpaCy for Named Entity Recognition (NER) and uses the mBART model to train and translate on the IWSLT 2017 English-to-Chinese translation dataset. The focus is on tagging named entities in the source language text to improve translation performance.

7.1 Data Processing

I preloaded the SpaCy **en_core_web_trf** model to identify named entities in the IWSLT 2017 English dataset. A custom function, **mark_entities**, was used to tag entities in the format «`{ent.label}:{ent.text}`>. Subsequently, another custom function, **preprocess_function**, was utilized to separate the English text and the Chinese translations (including **labels** and **entities** information) produced by **mark_entities**, assigning them to **inputs** and **targets**, respectively.

7.2 mBART50

Next, the `mbart-large-50-many-to-many-mmt` model and its tokenizer were loaded. The source language was set to `en_XX`, and the target language was set to `zh_CN`. The tokenizer encodes the data into a format suitable for the model, and in this project, it was configured with truncation and padding, with a maximum length defined as 256.

I split the **train_dataset** into 90% for training and 10% for validation, creating the **eval_dataset**. The validation set was used to evaluate the model's performance after each EPOCH.

7.3 Training Parameter Settings

For training parameter settings, I defined that validation would be performed at the end of every EPOCH, and the model would also be saved after each EPOCH. Since the training was conducted on Google Colab with an A100 GPU, these settings ensured that the model weights were not lost in the event of a disconnection.

In this project, the learning rate (`learning_rate`) was set to 1×10^{-5}, as it provides stability during fine-tuning. Both training and validation batch sizes were set to 4 to avoid GPU memory overflow. The regularization parameter (`weight_decay`) was set to 0.01 to prevent overfitting. The model was trained for 20 EPOCHs on the IWSLT training dataset, which contains 231,266 samples. The total training time exceeded 100 hours. During training, I used `logging_dir` to save logs for tracking progress.

7.4 Initializing the Trainer

The **mBART** model was loaded for training, along with the hyperparameters specified in **training_args**. The **train_dataset** and **eval_dataset** were loaded for training and validation, respectively, and the **MBart50Tokenizer** tokenizer was also included.

To prevent overfitting, I specifically configured **EarlyStoppingCallback(early_stopping_patience=3)**, which

stops training early if the model's performance on the validation set does not improve for 3 consecutive EPOCHs. Finally, the model weights were saved to the specified directory.

7.5 Model Training and Vaild Loss

Put the table .....

8 Eval Process

Talk about the Eval Process.....

9 Evaluation Results

9.1 Overall METEOR: 0.6649

METEOR (Metric for Evaluation of Translation with Explicit ORdering) is a translation evaluation metric that measures semantic similarity and word order consistency between the generated translations and the reference translations.

The METEOR score ranges from 0 to 1, with higher values indicating better performance. My mBART model achieved a METEOR score of 0.6649, demonstrating strong semantic similarity and good alignment with the reference translations.

9.2 Overall BERTScore

BERTScore is an evaluation metric based on embeddings from deep learning models such as BERT. It assesses translation quality from a semantic perspective by comparing the embeddings of the generated translations with the reference translations.

1. **Precision: 0.8008**
   Precision measures the proportion of content in the generated translation that correctly aligns with the reference translation. A score of 0.8008 indicates that 80.08% of the content in the generated translations accurately corresponds to the reference translations.

2. **Recall: 0.7933**
   Recall measures the proportion of the reference translation's semantic content that is correctly covered by the generated translation. A recall score of 0.7933 means that 79.33% of the reference translation's semantic content is successfully captured in the model's output.

3. **F1 Score: 0.7965**
   The F1 score is the harmonic mean of Precision and Recall, providing a balanced measure of translation accuracy and coverage. An F1 score of 0.7965 indicates that the model achieves a good balance between precision and recall.

9.3 Conclusion

These results indicate that the **EA-MT: Entity-Aware Machine Translation from English to Chinese** performs well in the translation task. Both the METEOR score (0.6649) and the BERTScore metrics (Precision: 0.8008, Recall: 0.7933, F1: 0.7965) are at a high level, reflecting strong semantic similarity and effective word order matching.

Although the Precision score is slightly higher than the Recall score, the balanced F1 score shows that the model excels in both generating accurate translations and covering the reference semantics. Overall, the EA-MT system demonstrates superior performance in English-to-Chinese machine translation.

10 Evaluation Metrics

We chose METEOR and BLEU for our evaluation metric, which assesses machine translation quality through unigram matching with human references. METEOR compares unigrams by surface form, stem, and meaning, calculating scores based on precision, recall, and fragmentation. And BLUE is One of the first metrics to correlate highly with human judgment, BLEU remains popular and cost-effective. However, BLEU does not account for intelligibility or grammatical correctness. Therefore, we use multiple evaluation methods to more truly reflect the translation capabilities of our model.

10.1 METEOR Calculation

METEOR (Metric for Evaluation of Translation with Explicit ORdering) is an automatic metric used for evaluating the quality of machine translation. Unlike BLEU, which focuses solely on precision, METEOR takes both precision and recall into account, providing a more balanced assessment of

translation quality. The key steps involved in calculating METEOR are as follows:

1. **Tokenization and Alignment:**
   - Tokenize the hypothesis (translated text) and reference (ground truth) sentences into words.
   - Align the tokens in the hypothesis with the tokens in the reference based on exact matches, stemmed matches, synonyms, or paraphrases (if applicable).
2. **Calculation of Precision and Recall:**
   - Precision (`P`): The fraction of tokens in the hypothesis that are aligned with tokens in the reference.

$$P = \frac{\text{Number of Matched Tokens}}{\text{Total Tokens in Hypothesis}}$$

   - Recall (`R`): The fraction of tokens in the reference that are aligned with tokens in the hypothesis.

$$R = \frac{\text{Number of Matched Tokens}}{\text{Total Tokens in Reference}}$$

3. **Harmonic Mean (F-score):** Combine precision and recall using a harmonic mean, which assigns greater weight to recall.

$$F_{\text{mean}} = \frac{(1+\beta^2) \cdot P \cdot R}{\beta^2 \cdot P + R}$$

   where $\beta = 3$, giving higher importance to recall.
4. **Fragmentation Penalty:** To penalize translations with disordered word sequences, a fragmentation penalty (`Pen`) is applied based on the number of contiguous segments in the alignment:

$$\text{Pen} = \gamma \cdot \left(\frac{\text{Chunks}}{\text{Matches}}\right)$$

   where γ is a predefined constant, typically set to 0.5.
5. **Final METEOR Score:** The final METEOR score is computed as:

$$\text{METEOR} = F_{\text{mean}} \cdot (1 - \text{Pen})$$

This metric is particularly well-suited for evaluating translations that emphasize semantic similarity and word order. By incorporating both precision and recall, along with penalties for disordered sequences, METEOR provides a robust evaluation of translation quality.

10.1.1 Semantic Similarity in METEOR for Non-English Languages

METEOR 中的語意相似的功能只能使用在英文當中, 所以我使用了哈工大的資料集建立了同義詞的詞林，Ⓕ也使用了 OpenHowNet 來計算語意的相似度

10.2 Semantic Similarity in METEOR for Non-English Languages

METEOR is designed to support semantic similarity calculations through the inclusion of stemming, synonyms, and paraphrasing. However, these features are primarily optimized for English and may not perform as effectively for other languages, such as Chinese. To address this limitation, I incorporated additional resources to enhance semantic similarity detection for Chinese translations:

- **Synonym Forest from HIT:** I utilized a synonym lexicon built from the Harbin Institute of Technology (HIT) dataset. This lexicon provides a comprehensive collection of Chinese synonyms, allowing the METEOR metric to match semantically equivalent words in translations. For example, synonyms such as 好 (good) and 优良 (excellent) can be aligned during evaluation.
- **Semantic Similarity with OpenHowNet:** To further improve semantic similarity detection, I integrated `OpenHowNet`, a large-scale knowledge graph specifically tailored for semantic understanding in Chinese. OpenHowNet provides detailed semantic representations for Chinese words, which were used to compute similarity scores between non-exact matches. This approach ensures that even if two words are not exact synonyms, their semantic similarity can still be quantified and considered in the evaluation.

- **Implementation in METEOR:** The synonym and semantic similarity components were incorporated into the alignment phase of METEOR. During the alignment process:
  - Exact matches between tokens were prioritized.
  - If no exact match was found, tokens were checked for synonym matches using the HIT lexicon.
  - If neither an exact nor a synonym match was found, OpenHowNet was used to calculate a semantic similarity score. Tokens with a similarity score above a predefined threshold (e.g., 0.8) were aligned.

This integration significantly enhanced the capability of METEOR to evaluate translations involving Chinese by capturing semantic equivalence more effectively. The combination of the HIT synonym lexicon and OpenHowNet ensured a robust semantic similarity mechanism, extending the functionality of METEOR beyond its native support for English.

10.3 BERTScore

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